

UCI: Division of Continuing Education
Accelerated Certificate Program (ACP): Data Science
Introduction to Python Programming
 I&C SCI_X426.58:

Homework#1

Date Given: April 3, 2024

Due Date: April 4, 2024

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There are 9 problems in this homework assignment. Write a program in Python to solve these problems.

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1. Assuming that variables a, b, c, d, and f are scalars, write Python statements to compute and display the following expressions. Test your statements for the following values.
 a = 1.12, b = 2.34, c = 0.72, d = 0.81, f = 19.83

Correct Answer

$$x = 1 + \frac{a}{b} + \frac{c}{f^2}$$

$$s = \frac{b-a}{d-c}$$

$$r = \frac{1}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}$$

$$y = ab \frac{1}{c} \frac{f^2}{2}$$

```
x = 1.4804634732516433
s = 13.555555555555541
r = 0.2535712749946248
y = 715.676598
```

2. Evaluate the following expressions in Python.

$$x = 5 + 8i$$

$$y = -6 + 7i$$

a) $u = x + y$

b) $v = xy$

c) $w = x / y$

d) $z = e^x$

e) $r = \sqrt{y}$

f) $s = xy^2$

Correct Answer

```
u = (-1+15j)
v = (-86-13j)
w = (0.3058823529411765-0.9764705882352942j)
z = (-21.594119667068036+146.83378286556217j)
r = (1.2687679963832803+2.758581560992251j)
s = (607-524j)
```

3. Use Python to compute the following expressions.

- a) $(3 + 6i)(-7 - 9i)$
- b) $\frac{5 + 4i}{5 - 4i}$
- c) $\frac{3}{2i}$

Correct Answer

```
a = (33-69j)
b = (0.21951219512195122+0.9756097560975611j)
c = -1.5j
```

4. Use Python NumPy to calculate the following expressions. Check your answers with a calculator. The log specified in this problem means log to the base 10.

- a) $e^{(-2.1)^3} + 3.47 \log(14) + \sqrt[4]{287}$
- b) $(3.4)^7 \log(14) + \sqrt[4]{287}$
- c) $\cos^2\left(\frac{4.12\pi}{6}\right)$
- d) $\cos\left(\frac{4.12\pi}{6}\right)^2$

Correct Answer

```
a = 8.09311298152669
b = 6023.964366415807
c = 0.3062422067739484
d = -0.05872703357180495
```

5. Suppose that $x = 6$. Find the results of the following operations by hand and use Python to check your results.

- a) $z = (x < 10)$
- b) $z = (x == 10)$
- c) $z = (x >= 4)$
- d) $z = (x != 7)$

Correct Answer

```
True
False
True
True
```

6. Find the results of the following operations by hand and use Python to check your results.

- a) $z = 6 > 3 + 8$
- b) $z = 6 + 3 > 8$
- c) $z = 4 > (2 + 9)$
- d) $z = (4 < 7) + 3$
- e) $z = 4 < 7 + 3$
- f) $z = (4 < 7) * 5$
- g) $z = 4 < (7 * 5)$
- h) $z = 2/5 >= 5$

Correct Answer

```
False
True
False
4
True
5
True
False
```

7. Suppose string 'A' is defined in a Python program as follows.

```
A = 'Data science professionals have promising career path'
```

Compute the following expressions.

- a. A[8]
- b. A[-1]
- c. A[-6]
- d. A[3:10]
- e. 'promising' in A
- f. A.lower().replace('career path', 'future').rstrip('d')
- g. A.index('science')

Correct Answer

```
a = e
b = h
c = r
d = a scien
e = True
f = ata science professionals have promising future
g = 5
```

8. Suppose in a Python program the following variables are defined.

```
X = 3          #integer
Y = '4'        #string
Z = '5.6'      #string
```

Compute the following expression: (str(X)+ Y + Z)

Correct Answer

```
answer = 345.6
```

9. Suppose in a Python program the following variables are defined.

```
X = '56'       #string
Y = '23.5'     #string
Compute: Z = str(int(X)+ float(Y))
```

Correct Answer

```
Z = 79.5
<class 'str'>
```